HALI SARAH PARSONS

Data Science | Bellevue University | Expected January 2027
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SUMMARY

Data science student (graduating January 2027) who turns messy data into insights businesses can act on. A year of project-based coursework in Python data wrangling, visualization, NLP, statistics, predictive modeling and business intelligence design. Comfortable owning a project from requirements to a finished deliverable.

DATA SCIENCE PROJECTS

Predicting Economic Damage from U.S. Storm Events — Group, Oct 2026

- Analyzed 410,311 NOAA storm events (2020–2025) to predict whether an event causes economic damage; Random Forest reached ROC-AUC 0.880 and average precision 0.669 vs. a 0.219 no-skill baseline.
- **My role: data visualization.** Built the charts comparing storm frequency with damage rates, showing common events (Heat, Hail) rarely carry recorded damage while Tornado and Strong Wind often do.

Custom Named Entity Recognition — May 2026

- Built a spaCy EntityRuler pipeline with four custom entity labels, extracting 2,501 entities from 2,540 sentences across five full-text PubMed Central articles (64,006 words) pulled live through an API.
- Applied sentence-level VADER sentiment to analyze how health outcomes and age groups are framed.

Digital Habits vs. Mental Health Analysis — May 2026

- Cleaned a 100,000-row dataset (removed 1,865 duplicates, flagged outliers with IQR, engineered sleep-quality categories) and built five visualizations on 98,135 records.
- Showed higher screen time tracks with lower mood across all three stress tiers.

BI Justification, Requirements & Architecture — BI coursework

- Wrote a business case (objectives, scope, timeline, budget, TCO, ROI) and a detailed requirements document with stakeholder interview inputs for a digital wellness analytics platform.
- Designed an end-to-end data architecture covering ingestion, ETL, data lake and warehouse, ML and BI layers.

Invoice Data Extraction in Power Automate — Apr 2026

- Trained and tested two custom AI Builder document-processing models (16 fields; 31 fields and 4 tables) and ran an automated flow successfully.

Additional Coursework Projects — 2026

- Sentiment analysis and logistic regression with TF-IDF, feature engineering and LDA topic modeling, Universal Sentence Encoder embeddings, and Bayes / Shannon entropy problem sets in Python.

EDUCATION

Bellevue University
Data Science
Expected January 2027

TECHNICAL SKILLS

LANGUAGES

Python, SQL, R / R Markdown, HTML, CSS, JavaScript

LIBRARIES

pandas, NumPy, SciPy, scikit-learn, spaCy, NLTK, matplotlib, seaborn

ANALYTICS & ML

Statistics (Bayes, Central Limit Theorem, entropy), regression, logistic regression, random forest, feature engineering, TF-IDF, topic modeling, named entity recognition, sentiment analysis, deep learning embeddings

BI & AUTOMATION

BI requirements and justification, data architecture and ETL design, Power Automate (AI Builder), Zapier, Jupyter / Colab, graphic design

COURSEWORK

- Essential Mathematics for Data Science
- Practical Python Data Wrangling
- Natural Language Processing & Text Analytics
- Data Visualization with R
- Business Intelligence Design
- Predictive Modeling Capstone